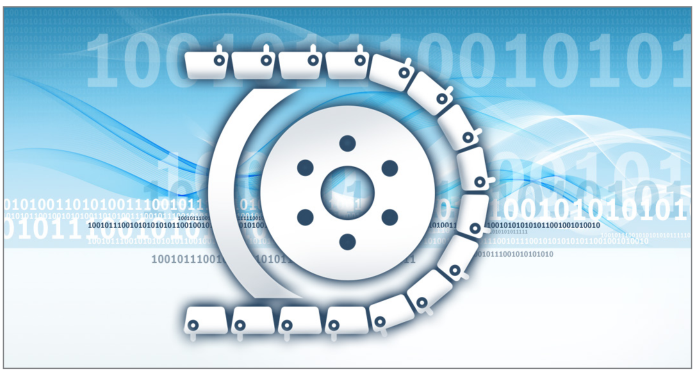
The Drive & Control Company

Rexroth
Bosch Group

Application software BODAS-drive DPC

Controllers RC12-10/30 and RC28-14/30 are phased out.
For new projects with this product, please contact DC-MH/SPL1

RE 95327
Edition: 10.2018



▶ Control solution for hydrostatic drivetrains for tracked vehicles
▶ Release 10

Features

- ▶ Combined drive and steering control for hydrostatic dual path tracked vehicles
- ▶ Proportional driving for precise and powerful travel and work operations
- ▶ Various steering patterns for user-oriented steering
- ▶ Closed-loop steering control for direct and accurate steering
- ▶ Multiple comfort functions like closed-loop speed control and automatic parking brake control for easy driving
- ▶ Diesel Hydraulic Control ECOdrive including engine speed control for improved efficiency and reduced noise emissions
- ▶ Additional functions like load limiting control and automatic fan control for best driver support
- ▶ Temperature derating for component protection
- ▶ Flexible interfaces – discrete or CAN
- ▶ Comprehensive diagnosis and configuration options via BODAS-service
- ▶ Integrated safety functions prepared compliant to EN ISO 13849 and ISO 25119
- ▶ Modular software concept for efficient customer-specific extensions
- ▶ Supporting process documents and tools for systematic integration into the machine
- ▶ BODAS-drive DPC is part of BODAS Bosch Rexroth design and application system for mobile electronics
- ▶ Software solution on Rexroth controller RC12-10/30

Contents

1 Introduction

BODAS-drive DPC BODAS-drive DPC is a software template written in BODAS-design Structured Text ST. The driving and steering control is realized in a BODAS-design library calc_DPC_V2.X.lib. It is a software solution for the Rexroth controllers RC12-10/30 or RC28-14/30 to control hydrostatic drivetrains of vehicles with tracks. The controlled drivetrain consists of an engine with or without CAN interface and a hydrostatic drive with two pumps and two hydraulic motors.

Energy efficiency with reduced fuel consumption and noise emissions can be reached with different eco modes. Prepared safety functions developed in accordance with the standards EN ISO 13849 and ISO 25119 are part of the software.

For more information about customer-specific extension, please consult your Bosch Rexroth contact.

2 Typical applications and variants

BODAS-drive DPC is designed to control the hydrostatic drivetrain of tracked mobile machines. Thanks to the multiple functions and configuration options it can be adapted to various applications like dozers, pavers, crawlers, compact track loaders or skid steer loaders.
BODAS-drive was developed using a generic approach and represents a safety element out of context. Assumptions have been made and documented accordingly for the applicable safety functions.

The following variants were tested in real applications, other variants are possible with changed parameter settings.

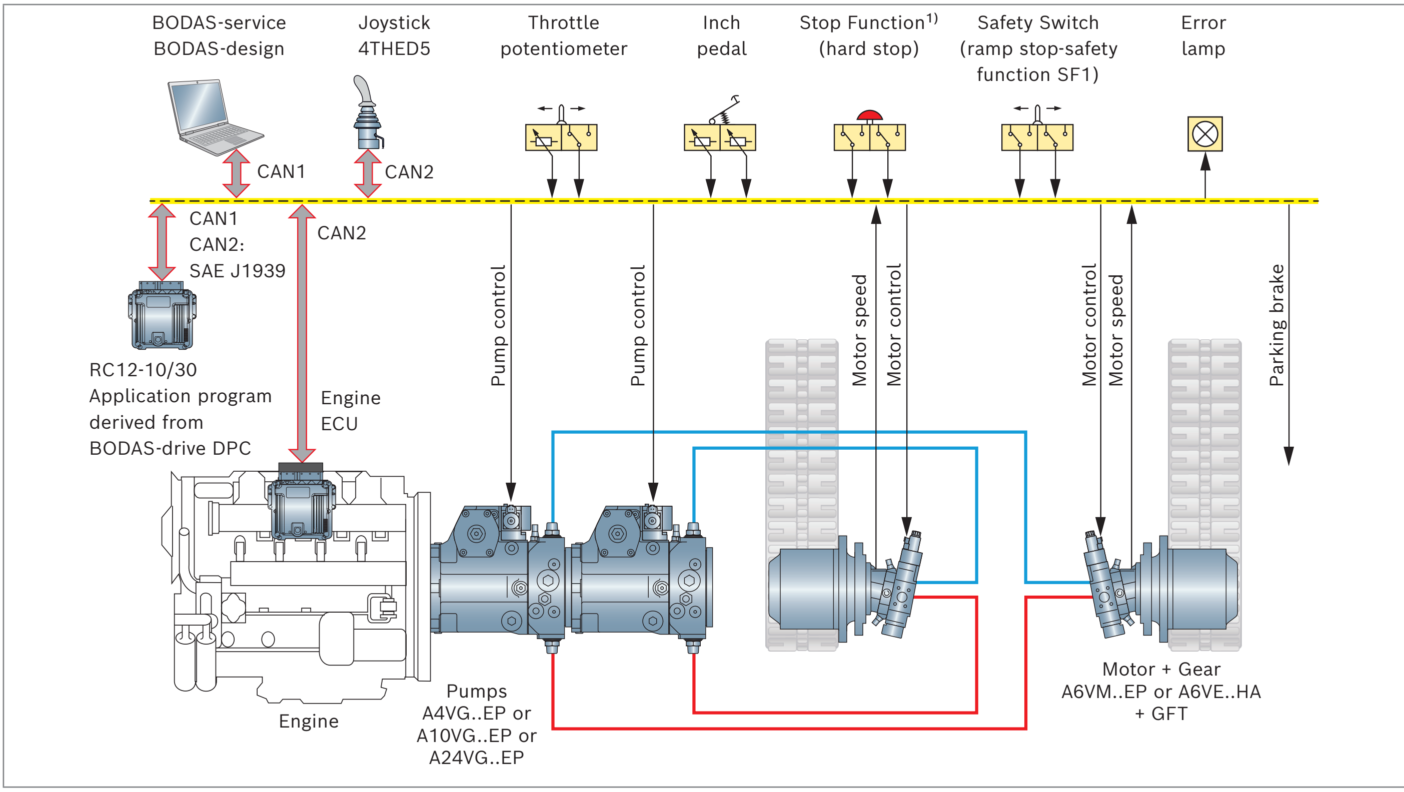
Notice
BODAS-drive DPC helps to realize functionality and safety on the machine level. The machine manufacturer must thoroughly check whether the functionality of BODAS-drive DPC can fulfil the requirements of the specific machine. If additional features are required, BODAS-drive DPC can be extended.

Vehicle	Dozer	Paver	Compact Track Loader CTL
Pattern	ISO_C curve	ISO_C curve	ISO_S curve integrated
Hydraulic System	EP/EP pump and motor control	EP/EP pump and motor control	ET pumps /MCR Motors
Voltage	12 V	24 V	12 V
Hydraulic motor speed sensor	DSM	HDD1	MCR
Engine type	CAN interface according J1939	CAN interface according J1939	CAN interface according J1939
Drive lever	HG joystick	Analog joystick	CAN joystick
Steer lever	HG joystick	Analog joystick	CAN joystick
Shift	no softshift, but electronically emulated gears	no softshift, but two electronically emulated gears	softshift MCR motors
Throttle	Potentiometer	no	no
Accelerator pedal	no	no	yes
Decelerator/Inch pedal	yes	yes, drive limit potentiometer	no
Fuel level measurement	no	yes	no
Mill mode	no	no	yes
Neutral valve	no	no	no
Implement hydraulics with own controller	no	no	yes
Fan control with hydraulics	no	yes	no

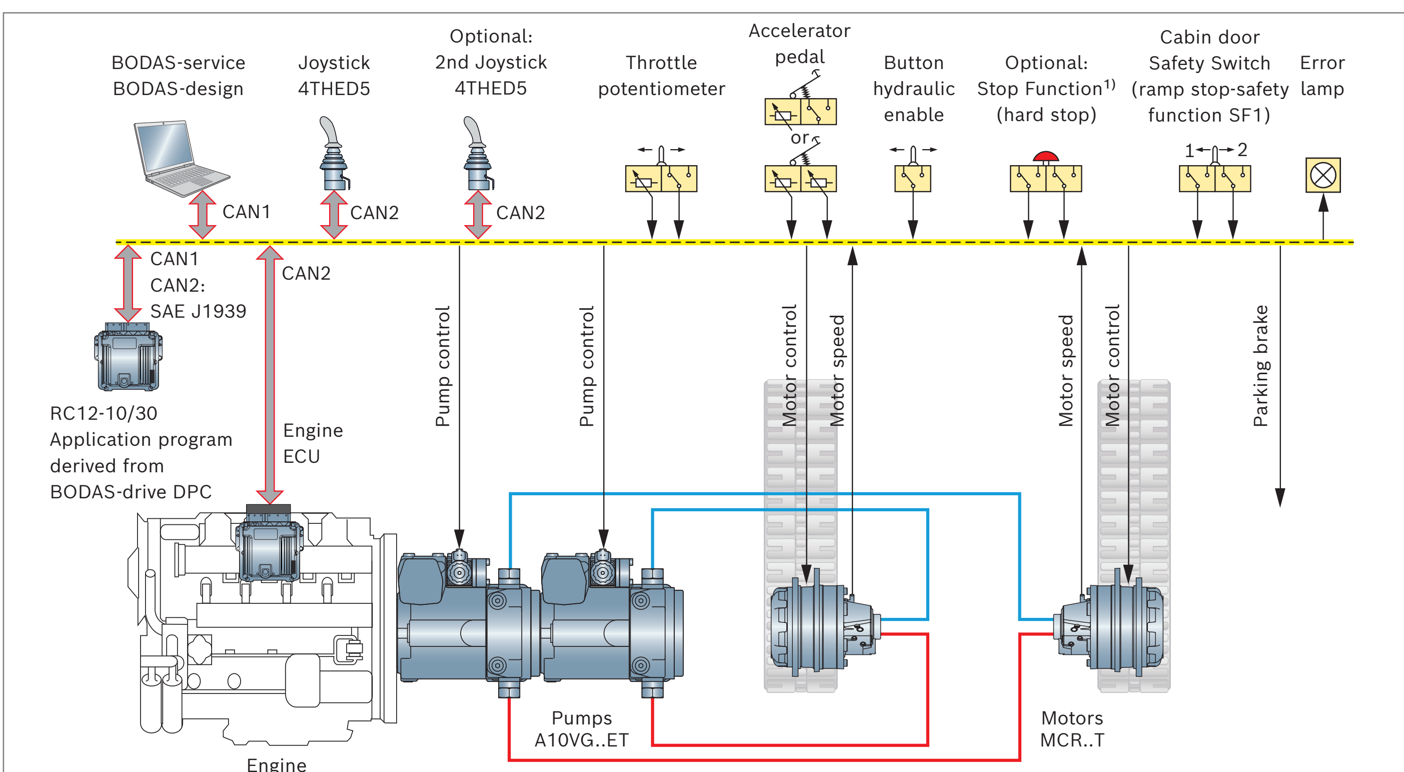
3 System description

3.1 System overview

▼ Example system overview for BODAS-drive DPC for dozers and pavers



▼ Example system overview for BODAS-drive DPC for compact track loaders



1) See Rexroth safety-relevant project planning instruction 95451-01-B

3.2 Drivetrain components

Engine
BODAS-drive DPC can be used with a diesel engine equipped with an ECU supporting CAN SAE J1939.
The engine ECU performs the complete control of the engine actuators and peripherals. A target speed request is set via BODAS-drive DPC. Actual engine values like speed or temperature are provided by the engine ECU. Important CAN messages are EEC1, ET1 and TSC1.
BODAS-drive DPC can be used with diesel engines without CAN SAE J1939 as well. Then a so called learning curve has to be recorded with BODAS-drive DPC. It stores for each throttle lever position the corresponding idle engine speed.

Drive pump
The A4VG is an axial piston pump in swashplate design with variable displacement and all components for a hydraulic closed circuit. The ET control module is a load sensitive control system. The output flow of ET pump is infinitely variable between 0 to 100%. Depending on the preselected current at solenoids **a** and **b** of the pressure-reducing valves, the stroke cylinder of the pump is proportionally supplied with control pressure. The pump displacement that arises at a certain control current is dependent on the speed and operating pressure of the pump. A different flow direction is associated with each pressure reducing valve.

The EP control module is a load independent control system. The output flow of the EP pump is proportional to solenoid current and the pump speed.

Drive motor
There is the possibility to apply fixed or variable hydraulic motors in the drivetrain. In case of a fixed motor the output speed of the hydraulic motor is proportional to the pump output flow. If the drivetrain is equipped with a variable axial piston motor A6VM with electric proportional control EP, the hydrostatic ratio can additionally be controlled by an electrically defined signal which sets the swivel angle of the motor. The closed loop EP control system ensures a constant swivel angle independent from the occurring high pressure.

4 Functional description

BODAS-drive DPC evaluates the input signals coming from the connected operator devices and machine sensors. Based on this, BODAS-drive DPC calculates the control values for the various actuators, such as the engine and hydrostatic drive. To map the BODAS-drive DPC functionality to a particular machine configuration, each function can be separately activated using BODAS-service. Thus only the relevant inputs are evaluated and just the needed outputs are activated. The diagnostic routines are adapted with regard to the activated functions.

Main functions	**Comfort functions**	**Energy efficiency and component protection functions**	**Configuration options**	**Safety functions**
See chapter 4.1	See chapter 4.2	See chapter 4.3	See chapter 4.4	See chapter 6.3 and 6.4
Driving control	Load limiting control	ECOdrive	Set vehicle type	Emergency stop
Steering control	Parking brake control	ECO auto idle	Set pattern type	Safe standstill
Speed controller (PID)	Inching	Temperature derating	Set hydraulic system	Safe deceleration
Steering controller (PID)	Milling mode	Diagnostics	Set battery voltage	Safe direction
Engine speed control	Three driving curves	Error Lamp	Set speed sensor type	Safe steering
Shift radial piston motors (soft shift)	Four steering curves		Set engine type	
	External steering		Set drive lever type	
	Fan control		Set steer lever type	
	Auto calibration		Set shift type	
	Fuel level detection		Set throttle type	
	Hand Throttle		Set inching type	
			Set mill type	
			Set fuel level type	

4.1 Main functions

Driving control

The driving (Y) axis of the joystick is read and one out of three selectable drive curve is used. For CAN joysticks a speed range (virtual gear) can be selected via a joystick button. The virtual gear can be selected independently or dependently on the drive direction. The inching is considered and also a reduction of the drive signal depending on the engine speed and hydraulic oil temperature.

Steering control

The steering (X) axis of the joystick is read and one out of four selectable steering curves is used.
Depending on the chosen pattern the steering signal is calculated.

Speed controller (PID)

If the speed controller is switched on, a closed loop controller (PID) controls the vehicle speed and tries to keep the actual vehicle speed according to the drive signal coming from the joystick.

Steering controller (PID)

The vehicle straight driving and the curve radius can be controlled by a closed loop (PID).

Engine speed control

The desired engine speed is calculated and send to the engine ECU via CAN SAE J1939 with TSC1.

4.2 Comfort functions

Load limiting control

This function provides protection against overloading and stalling of the engine. The actual and desired engine speed is monitored. If the actual engine speed drops too much, the hydrostatic ratio is reduced in order to reduce the load on the engine.
During a high load situation, the hydrostatic ratio has to be reduced rather quickly. There are two modes for the load limiting control: it can work on the pumps and the motors independently with two PID controllers, or one PID controller can work first on the motors and then on the pumps if the motors are on maximum displacement.

Parking brake control

The parking brake is opened if the joystick is not neutral. It is engaged after a waiting time set by a BODAS-service parameter if the vehicle comes to standstill.

Three driving curves

One out of three driving characteristics can be used defining the relation between drive lever angle and control for pumps and motors related to the driving. E.g. smooth, linear or progressive. Each characteristics could be changed by BODAS-service parameters.

Four steering curves

One out of four steering characteristics can be used defining the relation between drive lever angle and control for pumps and motors related to the steering. E.g. smooth, linear, fast, extremely fast. Each characteristics could be changed by BODAS-service parameters.

External steering

For the vehicle type PAVER the reading and using of an external analog signal coming from a sensor is prepared.

Fan control

An electronically controlled hydraulic fan drive is supported. The fan speed is controlled based on hydraulic oil temperature, engine coolant temperature and engine charge air temperature.

Auto calibration

There are two types of automatic calibrations of the hydrostatic drive, called grounded and lifted. This calibration should be started only if no persons are near the vehicle. Grounded calibration means the vehicle is standing on the ground and minimum currents of pump solenoids are calibrated and stored automatically.
The vehicle has to be moved up and tracks has to be in the air for the lifted calibration. It could be used for a hydraulic system with EP pumps and EP motors, EP pumps and constant motors and EP pumps and HA motors. This automated calibration finds the minimum and maximum currents for the pumps and motors.

4.3 Energy efficiency and component protection functions

ECOdrive

The ECOdrive can be activated with a switch. Then the nominal engine speed is reduced to a lower speed defined by a BODAS-service parameter. In the same ratio as the nominal engine speed is reduced from eco speed related to high idle the control signal for the hydrostat is increased for compensation. As long as the hydrostat signal does not reach the maximum, the vehicle speed is fully compensated and the vehicle is running with the same speed with and without eco mode.

ECO auto idle

There is an auto idle eco mode included. The nominal engine speed will be set to low idle if the joystick is set to the neutral position and a wait time, defined by a BODAS-service parameter, is over.

Temperature derating

If the hydraulic temperature exceeds a maximum temperature, the hydrostat control signal is reduced. The temperature and the derating level can be set as BODAS-service parameters.

Diagnostics

The state of the control unit as well as the connected devices are monitored during operation.
If a fault occurs, an appropriate error reaction is activated, such as power off, ramp stop or limp-home. Active and saved errors and process values are available via BODAS-service.

4.4 Configuration options

Set vehicle type

The vehicle type defines some aspects of the vehicle specific behavior. E.g. if the type PAVER is set, a change from forward to backward driving is done via the state neutral but for the type DOZER or CTL directly from forward to backward. For the type PAVER steering is possible from an external sensor signal. For the type PAVER the total distance is calculated based on the speed signal. Additionally the relationship between vehicle type and each function can be modified.

Set pattern type

One joystick is used for to command drive and steer, called XY joystick. This joystick is located normally on left hand side of the driver seat e.g. for a dozer.
The pattern type defines the relation between XY joystick lever position and drivetrain behavior of the two hydraulic circuits for left and right track.
With the pattern type “C-curve” it reacts like in a car. If the drive joystick is set forward right, the vehicle moves to forward right direction. If the joystick is set to backward right, the vehicle moves to backward right. This is common for dozers, pavers.
With the pattern type “S-curve” it is the opposite. If the drive joystick is set forward right, the vehicle moves to forward right direction. If the joystick is set to backward right, the vehicle moves to backward left.
With the pattern type “S-curve integrated” the behavior is the same as for “S-curve”. Additional the change between driving and counter rotation is fluently, and reacts like a hydraulic joystick. This is common for compact track loaders CTL and skid steer loaders SSL.

Set hydraulic system

The following hydraulic systems for the drivetrain could be chosen: representing pump and motor combinations: EP pumps with EP motors, EP pumps with HA motors, EP pumps with constant motors, ET pumps with MCR motors.

Set battery voltage

12 V or 24 V is available.

Set speed sensor type

These hydraulic motor speed sensor types are available: HDD1, DSA1, HDD2, DSA2, MCR with transient protection (Material number 854983), DSM sensor.

Set engine type

It can be defined whether the diesel engine ECU has a CAN SAE J1939 control or not.

Set drive lever type

There is an XY joystick usual depending on the vehicle type, e.g. for dozers. The X axis is used to command the steering signal and the Y axis the driving signal.
On pavers, there is a separate joystick for the driving common, with one axis only. Then a second element is necessary to command the steering. That is the reason for splitting drive lever type and steer lever type.
The following joystick types are possible for driving:

- CAN joystick on CAN_1
- HG 405 CAN/analog joystick on CAN_1 (see data sheet 95241)
- Bosch Rexroth THED5-CAN joystick on CAN_2 (see data sheet 29696)
- CAN SAE J1939 joystick called BJM1 on CAN_2
- Joystick with two analog outputs with cross characteristics (e.g. one from 0.5 V to 4.5 V, the second from 4.5 V to 0.5 V)

Set steer lever type

In one option there is a separate steering lever supported. There is an additional setting just for the steering lever possible due to the fact that on some vehicles like pavers there are separate levers for driving and steering.
The following joystick types are possible for driving:

- CAN joystick on CAN_1
- HG 405 CAN/analog joystick on CAN_1 (see data sheet 95241)
- Bosch Rexroth THED5-CAN joystick on CAN_2 (see data sheet 29696)
- CAN SAE J1939 joystick called BJM1 on CAN_2
- Joystick with two analog outputs with cross characteristics (e.g. one from 0.5 V to 4.5 V, the second from 4.5 V to 0.5 V)

Set shift type and Soft shift

For a hydraulic system with ET pumps and MCR motors soft shift is available. The jumps of the motor displacements during switching from first to second gear and second gear to first gear will be compensated by pumps displacement. Two transmission ratios are supported: 100% and 50%. Changing from one ratio to another is possible during standstill and during driving manually. Pressing the shift up button or shift down button.

Set throttle type

The electrical interface of the throttle lever can be set to a potentiometer with a high idle switch or two analog outputs with cross characteristics (e.g. one from 0.5 V to 4.5 V, the second from 4.5 V to 0.5 V). Also no throttle lever can be chosen, then the signal for the target engine speed is coming from another source, example a display.

Set inching type and Inching

Using an inch pedal the pump and motor control can be reduced independently of the joystick lever position. There are several types of inching possible e.g. normal inching for dozers, the engine speed could also be reduced with the inch pedal or the parking brake could not be engaged if the inch pedal is fully depressed. Another inch mode works reverse, common on pavers: If the inch lever is on maximum, then maximum vehicle speed is possible and can be reduced with smaller inch lever positions.

Set mill type and mill mode

Two different mill modes are possible for the compact track loader: the vehicle speed during milling could be controlled with a mill mode speed potentiometer or with CAN joystick buttons up and down.

Set fuel level type and Fuel level detection

The fuel level could be measured if there is a fuel level sensor mounted. The measured voltage is converted into a percentage of fuel level with a set of BODAS-service parameters.

5 Electrical interfaces

5.1 Maximum output currents

The maximum allowed current per output pin is individually indicated in the connection diagram. Within the ECU, one output stage drives the current for two output pins for EP motors. The current sum of both pins must remain below the maximum allowed current of the output stage.
Check maximum flowing current for digital outputs, especially for the brake output:

- At highest voltage 12 V to 14.5 V, 24 V to 27 V diveded by minimum solenoid resistance at coldest temperature. This current has to be smaller than allowed current for the pin.

For additional information, also refer to Rexroth data sheet 95204 for BODAS controller RC series 30.

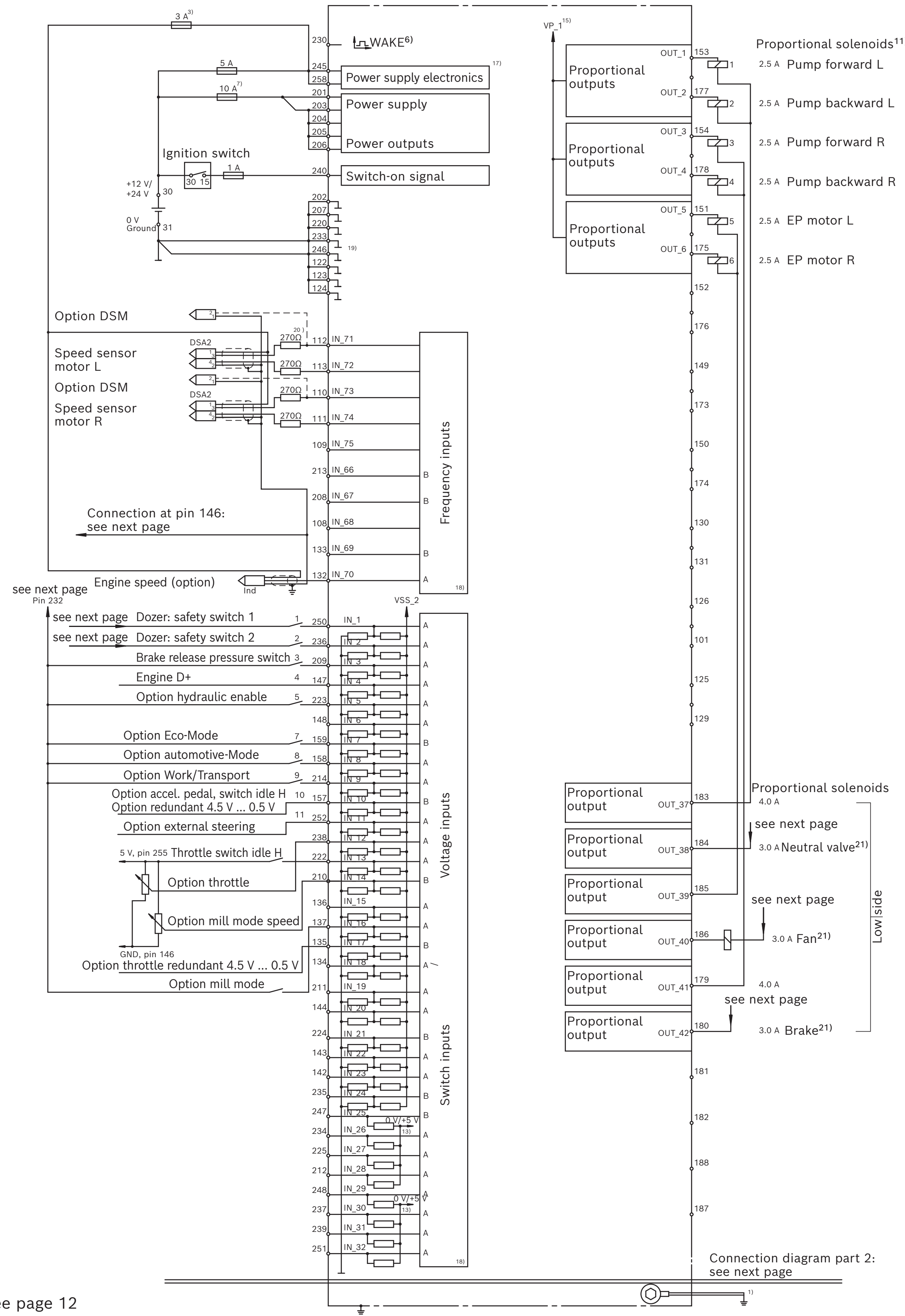
5.2 CAN signals

▼ Up to four CAN channels are supported:

CAN 1	250 kBaud	Communication with CAN joystick, BODAS-service or BODAS-design
CAN 2	250 kBaud	J1939 Standard. Communication with diesel engine and other ECUs and CAN joystick
CAN 3	250 kBaud	Communication with Virtual Test Box in CANalyzer
CAN 4	1000 kBaud	Communication via CAN calibration protocol (CCP). Supporting development tools like CANape, INCA or equivalent for advanced measuring options. **Caution:** CCP must not be used for parameter setting or calibration.

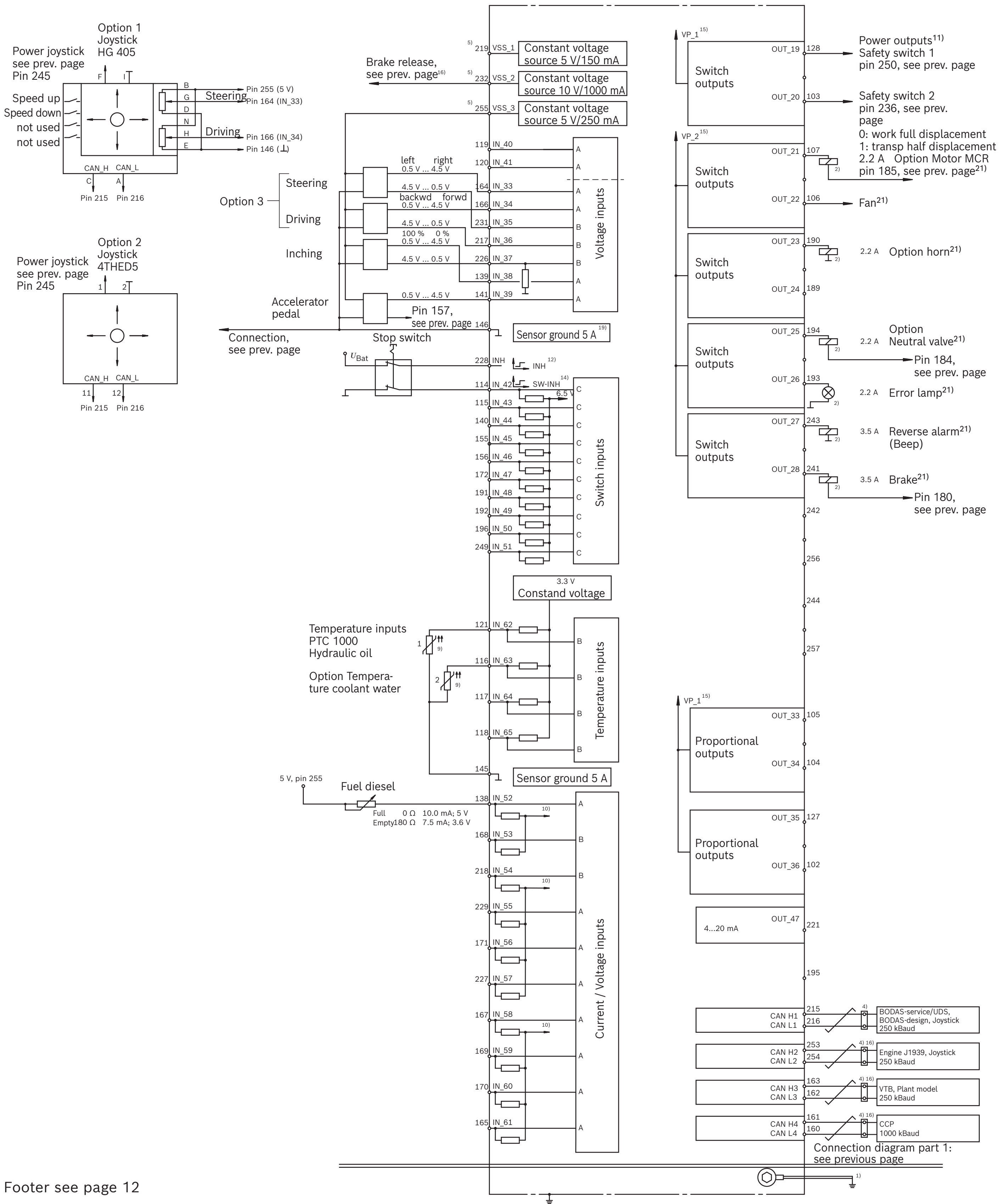
For details, see the CAN database, which is part of the BODAS-drive DPC documents and tools container.

5.3 Connection diagram RC12-10/30
Part 1



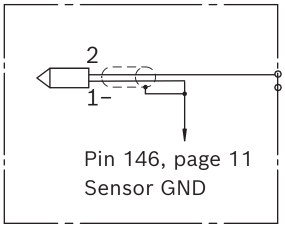
Footer see page 12

Connection diagram RC12-10/30
Part 2

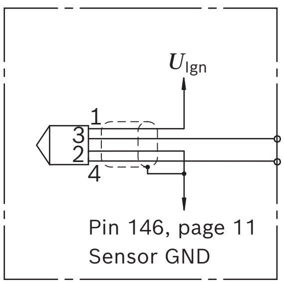


Footer see page 12

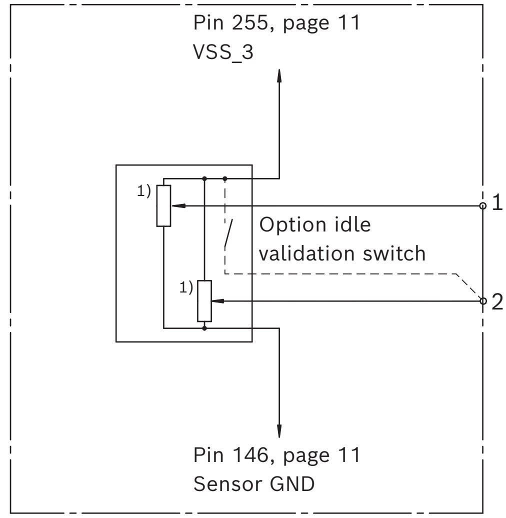
▼ **Speed sensor DSM**



▼ **Speed sensor HDD2 (NPN only), DSA2**



▼ **Connection of a dual channel potentiometer**



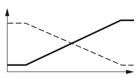
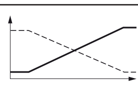
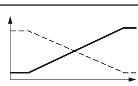
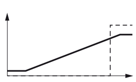
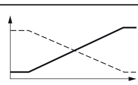
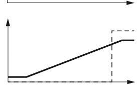
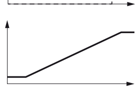
1) Active or passive potentiometer (1-4k); signal range can be calibrated with BODAS-service

Footer from page 10 and 11

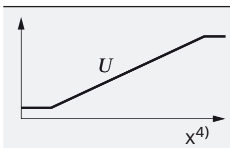
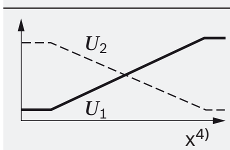
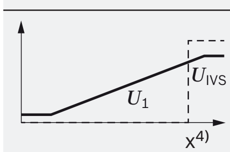
1) Short, low-resistance connection from a case screw to the vehicle ground.
2) Separate ground connection to battery (chassis possible).
3) Separate fuses for switches and sensors necessary. Sensor supply application specific.
4) CAN-Bus: termination resistor 120 Ω and twisted pair wire necessary.
5) Outputs 5 V/10 V can also be used as sensor supply alternatively.
6) Temporary wake up of the controller when a signal >8 V is applied for more than 1 sec.
7) Note max. current consumption with simultaneous actuation of proportional solenoid and switched outputs.
8) Separate ground connection for current source to battery, controller GND possible.
9) Can be used as switch inputs if externally switched to GND.
10) For use as voltage inputs (0...10 V), the load can be stiched by the software in groups for these inputs.
Groups: inputs 1...2, inputs 3...6, inputs 7...10
11) Outputs arranged in groups, each with 2 output stages. Maximum permissible output current of a group: 5 A.
12) Primary deactivation channel for proportional and switch outputs: enabling with level >4.5 V, deactivation with level <1 V, cable break leads to deactivation.
13) Input groups may be switched to pull down or pull up in software.
14) Secondary deactivation channel for proportional and switch outputs: enabling with level <0.8 V, deactivation with level >1.7 V, cable break leads to deactivation.
15) Supply can be switched by the software.
16) Is switched off when the watchdog triggers. Is switched off shortly for diagnosis purposes when a main switch is initially activated.
17) If power is disconnected during operation no data can be saved to non-volatile memory and no after-run.
18) A and B indicate different A/D converters which may be selected for redundancy reasons.
19) Terminal 31 (supply ground) and sensor ground are bridged at a star point in the control unit and connected to the housing.
20) 270 Ω recommended for DSA1 BR10 sensors (HDD npn, DSA BR12, DSM without resistors).
21) Check maximum flowing current = highest voltage 12 V → 14.5 V → 27 V divided by minimum resistance at coldest temperature. It has to be smaller than maximum allowed current for this pin.

5.4 Inputs

Range inputs

	Discrete				CAN Bus	
	Supported electrical interfaces	**Maximum signal voltage range**[1)]	**Open input voltage V**[2)]	**Remarks**	**Supported CAN messages**	**Remarks**
Drive lever				Two channels per joystick: ▶ Positive and negative deflection of x and y axis ▶ Signal range of each channel and neutral position can be learned via trimming functionality of BODAS-service	**CAN** SAE J1939 Proprietary	Proprietary messages include checksum and message counter
Steer lever						
Inch pedal[3)]			0 V		–	
Accelerator pedal	 	0.5 to 4.5 V			–	
Hand throttle	  		2.7 V	▶ Idle validation switch must be connected to 5 V	–	

▼ Supported electrical interfaces

	This interface expects one analog voltage signal in the range from 0.5 V to 4.5 V. The details of the signal characteristics including start and end point as well as the allowed tolerances can be configured via parameters.
	This interface expects two opposing voltage signals in the range from 0.5 V to 4.5 V. The first signal is the leading signal and the second signal is used for plausibility check. The details of the signal characteristics including start and end point as well as the allowed tolerances can be configured via parameters.
	This interface expects one voltage signal in the range from 0.5 V to 4.5 V and one on/off idle validation switch U_{IVS} signal. The voltage signal is the leading signal and the on/off signal is used as a plausibility check.

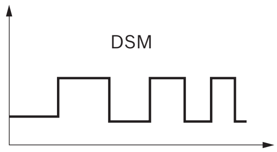
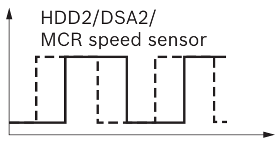
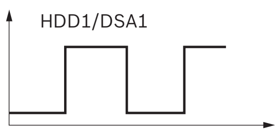
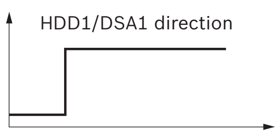
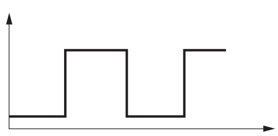
1) Passive sensors such as potentiometers must be connected to a 5 V sensor supply (VSS_1 or VSS_3). Active sensors must be supplied as specified by sensor data sheet (VSS_1, VSS_2, VSS_3 or U_{Ign}).

2) Voltage measured in case of unconnected signal pin. Voltage results from internal circuitry of RC.

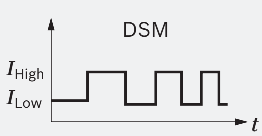
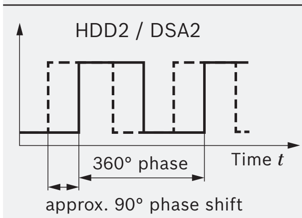
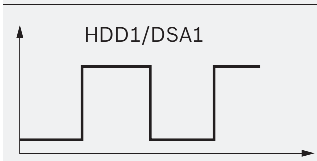
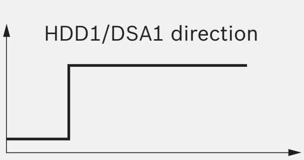
3) A combined brake / inch pedal must be connected as inch pedal

4) x = angle α or line S

Speed sensor inputs

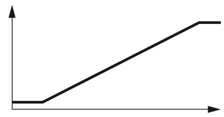
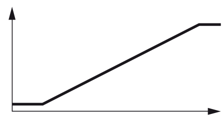
	Discrete			CAN Bus	
	Supported electrical interfaces	**Sensor supply**[1]	**Remarks**	**Supported CAN messages**	**Remarks**
Left / right hydraulic motor speed		Signal channel of RC	Supported positions and combinations of speed sensors are shown below	–	
		Speed sensor U_{Ign}			
	 	U_{Ign}		–	
Engine speed		–	Optional	**CAN** SAE J1939 Standard	Standard EEC1 message

▼ Supported electrical interfaces

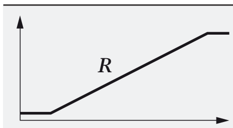
	DSM Sensor: This interface expects one frequency with coded error and direction information.
	HDD2[2] / DSA2 Sensor: This interface expects two frequencies with direction-dependent phase shift.
 	HDD1 / DSA1 Sensor: This interface expects one frequency and a digital direction signal.

1) Sensor dependent supply voltage
2) Only NPN type supported

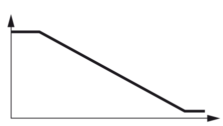
Temperature inputs

	Discrete		CAN Bus	
	Supported electrical interfaces	**Remarks**	**Supported CAN messages**	**Remarks**
Hydraulic motor oil temperature		Supported sensor: TSF	–	
Engine coolant temperature		Optional: Supported sensor: TSF	**CAN** SAE J1939 Standard	Standard ET1 message

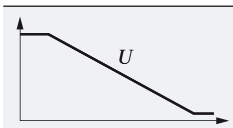
▼ Supported electrical interfaces

	This interface expects a temperature-dependent sensor resistance. The available measurement range of the RC is supported by the software.

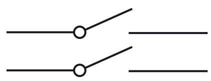
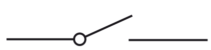
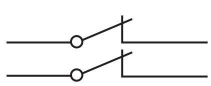
Fuel level inputs

	Discrete	
	Supported electrical interfaces	**Remarks**
Fuel level sensor		Optional For example with these values: 5 V, 0 Ω: full 3.6 V, 180 Ω: empty The voltage for empty, 10%, 20%...100% can be configured in 11 steps via BODAS-service.

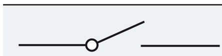
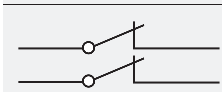
▼ Supported electrical interfaces

	This interface expects a fuel level-dependent sensor resistance. The available measurement range of the RC is supported by the software.

Switch inputs

	Discrete			
	Supported electrical interfaces	**Closed[1] switch voltage**	**Open[2] switch voltage**	**Remarks**
Safety switch 1 **Safety switch 2**		Fed by outputs	2.7 V	With build in diagnostic test
Brake release **Pressure switch**		10 V	2.7 V	Optional
Engine D+ **(generator voltage)**		U_{Ign}	2.7 V	Optional
Hydraulic enable		10 V	2.7 V	For CTL, SSL
Eco mode				
Automotive mode				
Transport				
External steering				For Pavers optional
Mill mode				For CTL, SSL optional
Emergency stop switch		0 V and U_{Bat}	0 V and 6.5 V	Switch must have two normally closed contacts

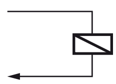
▼ **Supported electrical interfaces**

	This interface expects an on/off signal coming from a push button or switch with normally open contact.
	This interface expects two redundant on/off signals coming from a push button or switch with two normally closed contacts.

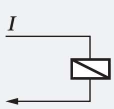
1) External potential connected to RC by a closed switch.
2) Potential measured at RC pin at open switch.
Voltage results from internal circuitry of RC.

5.5 Outputs

Proportional outputs

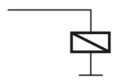
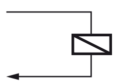
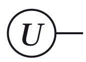
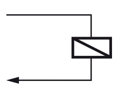
	Discrete				CAN Bus	
	Supported electrical interfaces	**Default output logic**[1] (without current)	**Invertible**[2]	**Remarks**	**Supported CAN messages**	**Remarks**
Pump solenoids		Pump swivels to V_{g0}	no	Pump forward and backward left is connected to one lowside switch. Pump forward and backward right is connected to one lowside switch.	–	
Hydraulic motor solenoids		Motor swivels to $V_{g\ max}$	no	Both hydraulic motor solenoids are connected to one low-side switch.		
Fan			no	Feeded by digital output. Not realized as safety function.	–	
Desired engine speed					**CAN** SAE J1939 Standard	Standard TSC1 message

▼ Supported electrical interfaces

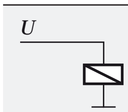
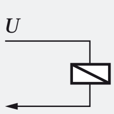
	This high-side output expects a solenoid connected to a low-side switch. The low-side switch is an additional safety path for switching off the output in case of external short circuits. Up to four high-side switches can be connected to one low-side switch. The proportional output current is generated via PWM closed-loop control. The details of the solenoid characteristics including minimum and maximum resistance can be configured via parameters.

1) For output that is shut off (no current), expected default machine behavior for a deactivated output (no current driven by high-side output)
2) Indicates if output logic can be inverted by parameter current

Switch outputs

	Discrete			
	Supported electrical interfaces	**Default output logic**[1]	**Invertible**[2]	**Remarks**
Fault lamp		Lamp is off	no	
Horn				Optional
Reverse alarm				Optional
Parking brake valve		Parking brake is open	yes	Parking brake valve solenoid is connected to one low-side switch
Pump neutral valve		No short circuit between pump X1, X2	yes	Pump neutral valve solenoid is connected to one low-side switch
Supply of safety switch 1		Supply is on after start-up	no	With build in diagnostic test
Supply of safety switch 2				
Supply of fan solenoid				

▼ **Supported electrical interfaces**

	This high-side output expects a solenoid connected to ground potential. The output voltage is switched to battery voltage or ground. The details of the solenoid characteristics including minimum and maximum resistance can be configured via parameters.
	This high-side output expects a solenoid connected to a low-side switch. The low-side switch is an additional safety path for switching off the output in case of external short circuits. Up to four high-side switches can be connected to one low-side switch. The output voltage U is switched to battery voltage or ground. The details of the solenoid characteristics including minimum and maximum resistance can be configured via parameters.
	This high-side output expects a resistance (e.g. a lamp) connected to ground potential. The output voltage is switched to battery voltage or ground. The details of the resistance characteristics including minimum and maximum resistance can be configured via parameters.

1) For output that is shut off (no current), expected default machine behavior for a deactivated output (no current driven by high-side output)
2) Indicates if output logic can be inverted by parameter current

5.6 Power supplies

Battery power supply

- 12 V and 24 V batteries are supported.
- Different solenoids are to be used depending on the battery voltage. Thus different load resistances are expected for error detection. The RC controller is using an after-run functionality. Therefore, battery power supply must not be disconnected within a time period of 2 seconds after switching of ignition.

Sensor supplies

U_{Bat}

- This potential is connected to battery voltage and is protected by a 5 A fuse.
- It is used for power supply of ECU electronics, the emergency stop switch and for sensors requiring battery voltage as power supply.

U_{Ign}

- This potential is connected to the ignition switch and is protected by a 1 A fuse.

VSS_1

- This potential is connected to a 5 V constant voltage sources supplied by the RC.
- It is not used in the BODAS-drive DPC wiring harness.

VSS_2

- This potential is connected to a 10 V constant voltage source supplied by the RC.
- It is used for some switches.

VSS_3

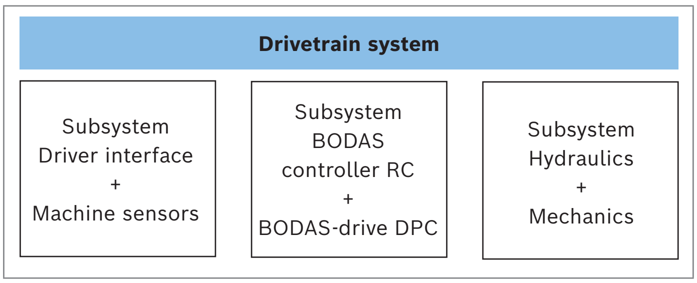
- This potential is connected to a 5 V constant voltage source supplied by the RC.
- It is used for sensors requiring a 5 V power supply and for potentiometers.

6 Functional safety in accordance with EN ISO 13849 and ISO 25119

For general information see the Rexroth brochure 08511: “10 Steps to Performance Level”.

6.1 Approach

BODAS-drive uses a subsystem approach as stated in the figure below.



The machine manufacturer can use the BODAS controller RC + BODAS-drive DPC subsystem within the machine safety design to realize safety functions for the drivetrain system. The described characteristics within this data sheet refer to the BODAS controller RC + BODAS-drive DPC subsystem.
Prepared safety functions developed according to the standards EN ISO 13849 and ISO 25119 are part of the software. If the performance level for the BODAS-drive DPC subsystem is sufficient to reach the required overall performance level, BODAS-drive can be used based on the customer-specific risk assessment. In any case, the requirements of the relevant safety standard must be fulfilled at the drivetrain and machine level.

6.2 Concept

The software of the BODAS controller RC + BODAS-drive DPC subsystem utilizes an inherent safety concept. This means that all noted safety functions are realized not by dedicated monitoring software, but by a safe implementation of the safety relevant software program parts. The BODAS-drive DPC software has been developed according to PL d and SRL 1. The software comes with 6 predefined safety functions, but is not limited to these. For additional safety functions, contact your Bosch Rexroth sales partner.

This inherent approach is combined with a safe diagnosis system within the BODAS-drive DPC software, fulfilling the requirements of category 2: Detection of and appropriate reaction to hardware faults.
Additional test equipments (TE’s) are included for the safety functions to detect faults on machine level. The monitoring with these TE’s could be switched on or off in the BODAS-drive DPC software and is recommended for PL c and necessary for PL d.

As BODAS-drive represents a safety element out of context, the machine manufacturer must verify whether it is the right product for the specific application. In any case, the machine manufacturer is responsible to fulfill the overall safety requirements at the machine level.

Two example safety concepts are part of the BODAS-drive DPC delivery: one for compact track loader and one for dozer and paver.

The following approach shows how the machine manufacturer can reach the required performance level for a specific application applying BODAS-drive.

1. Risk assessment

To determine all safety requiremts it is strongly recommended to observe the following procedure.

- Consideration of application specific standards for the relevant machine, if applicable
- Performance of a risk assessment
- Identification of the safety functions
- Determination of the required performance level (PLr) for each safety function

2. Safety concept

- Comparison of risk assessment results with safety functions offered within BODAS-drive.
- If the safety requirements of the application cannot be fulfilled with the existing safety functions of BODAS-drive DPC, the product must not be used. In this case, consult your Bosch Rexroth contact regarding a customized solution before the next step can be done.
- Developing a safety concept for the complete machine and drivetrain
  - Applying BODAS-drive inherent safety approach
  - Using and adaption of safety function test equipments (TE) if necessary for PL c and PL d
  - Creating block diagrams
  - Calculation of the overall performance level. The SISTEMA tool from IFA may be used. Calculation examples are provided for all safety functions. Once the safety function groups and the characteristics of the sensors and actuators have been provided, it can directly calculate the probability of failure per hour and the performance level achieved.

3. Integration and parameterization of BODAS-drive DPC in the machine

- Integration of BODAS-drive DPC in the machine environment interfacing the wiring harness and devices which are selected according to the safety requirements.
- Setting parameters with BODAS-service according to the application-specific requirements.

4. Validation

- Generation of an appropriate application-specific approval test specification. The BODAS-drive approval test specification is part of the documents and tools container and can be used as a starting point. An application-specific adaptation is required in any case.
- Execution of approval tests specific to application and project as well as documentation of the results.

6.3 Safety functions for dozer, paver

The software has been developed to fulfil the requirements of PL d (DIN EN ISO 13849) and SRL 1 (ISO 25119). The ECU hardware fulfils the requirements of category 2 according to DIN EN ISO 13849 and ISO 25119.

Relationship between safety functions and vehicle functions

The following tables show the dependencies between safety functions and configured functionality.
The following table shows, which functionality is related to each declared safety function. For example, to know which functionality has a dependency for e.g. the safety function “Safe Standstill”, the “x” marks in the line of “Safe Standstill” show which functionality is affected.

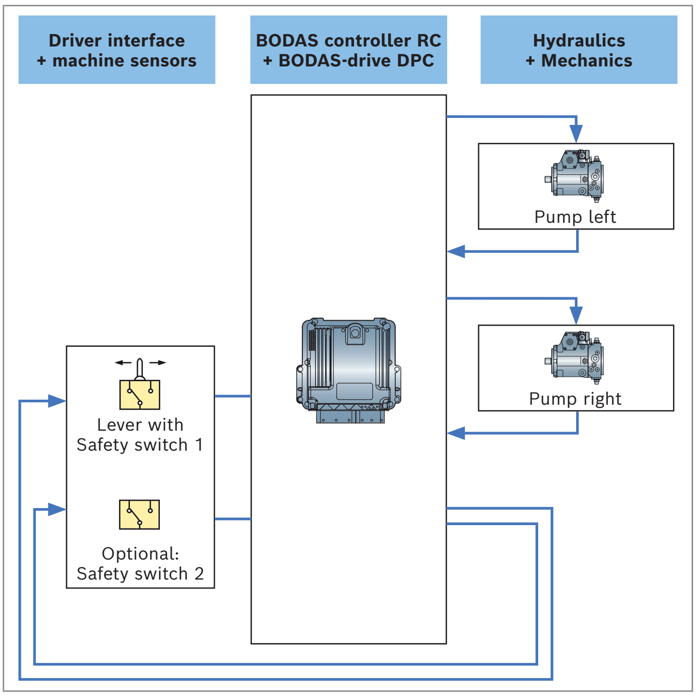
Safety Functions

Function	SF1 Emergency stop[1]	SF2 Safe standstill	SF3 Safe deceleration	SF4 Safe direction	SF5 Safe steering
in_drive: in_analog_drive (F2, F7)		x	x	x	x
in_drive: in_analog_inch (F9, F15)		x	x	x	x
in_drive: in_analog_steer (F2, F8)		x	x	x	x
in_CAN_Joystick: in_CAN_drive (F2, F7)		x	x	x	x
in_CAN_Joystick: in_CAN_steer (F2, F8)		x	x	x	x
in_Engine: in_accelerator (F13)			x		
in_Engine: in_throttle (F12)			x		
calc_Engine: calc_nominal_speed (F12)			x		
calc_Engine: calc_pump_limit (F9, F15)			x		
in_drive, action: in_switches (F9, F15)	x				
err_RC	x				
err_RC: error_reaction	x				
lib: state_and_drive_ramp (F9, F15)	x	x	x	x	x
lib: speed_controller (F17)	x	x	x	x	x
lib: steering (F10)	x	x	x	x	x
lib: steering_controller_LR (F10)	x	x	x	x	x
lib: steering controller separat (F10)	x	x	x	x	x
lib: pumps (F9, F15)	x	x	x	x	x
lib: brake (F9, F15)	x	x	x	x	x
out_drive (F4)	x	x	x	x	x

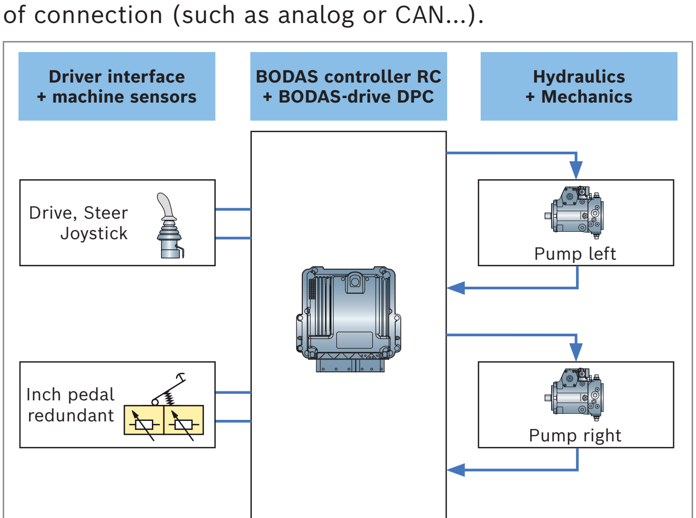
Relationship between functions and safety functions for paver, dozer
F: Function numbers, see detailed software documentation
lib: calc_DPC_VX.X.lib: calc_drive

1) Activated via “Safety Switch”, see system overview chapter 3.1

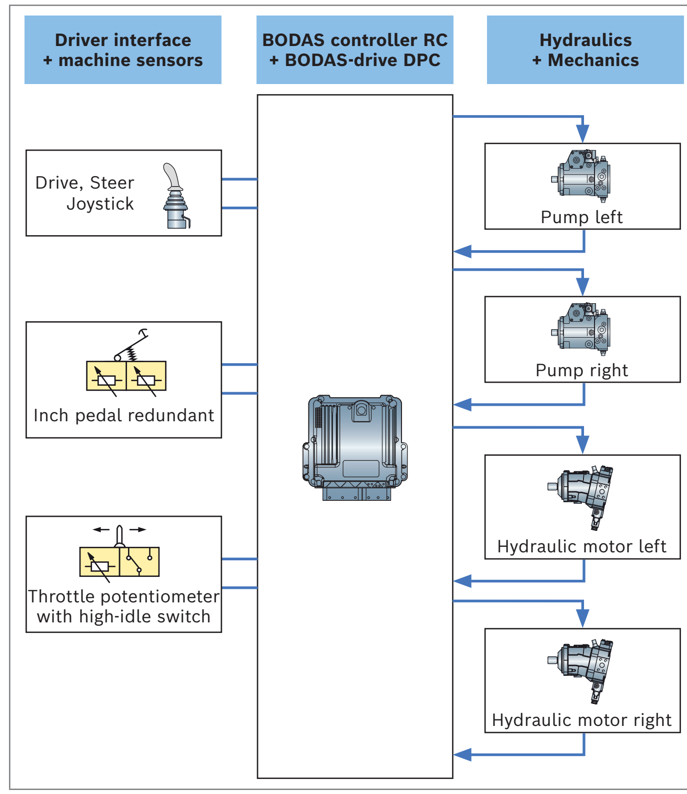
SF1: Emergency stop
This safety function stops the movement and inhibit vehicle movement from standstill if the safety lever is set down. The driver can be outside cabin.
PFH value for the BODAS controller RC + BODAS-drive DPC subsystem depends on the connected sensors for the safety switch(es).



SF2: Safe standstill
This safety function ensures no unwanted self-propelled movement of the machine caused by an unwanted torque of the hydrostatic transmission in standstill condition if there is no driver command.
PFH value for the BODAS controller RC + BODAS-drive DPC subsystem depends on the connected sensors and the type of connection (such as analog or CAN...).

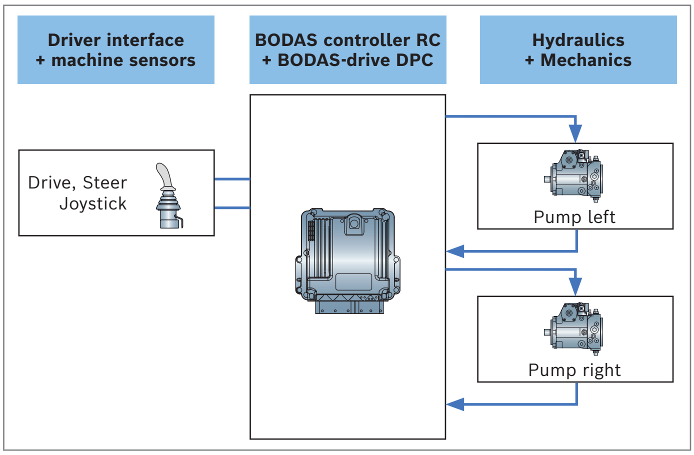


SF3: Safe deceleration
This safety function ensures that the hydrostatic transmission issues a braking torque when the driver demands it.
PFH value for the BODAS controller RC + BODAS-drive DPC subsystem depends on the connected sensors and the type of connection (such as analog or CAN...).



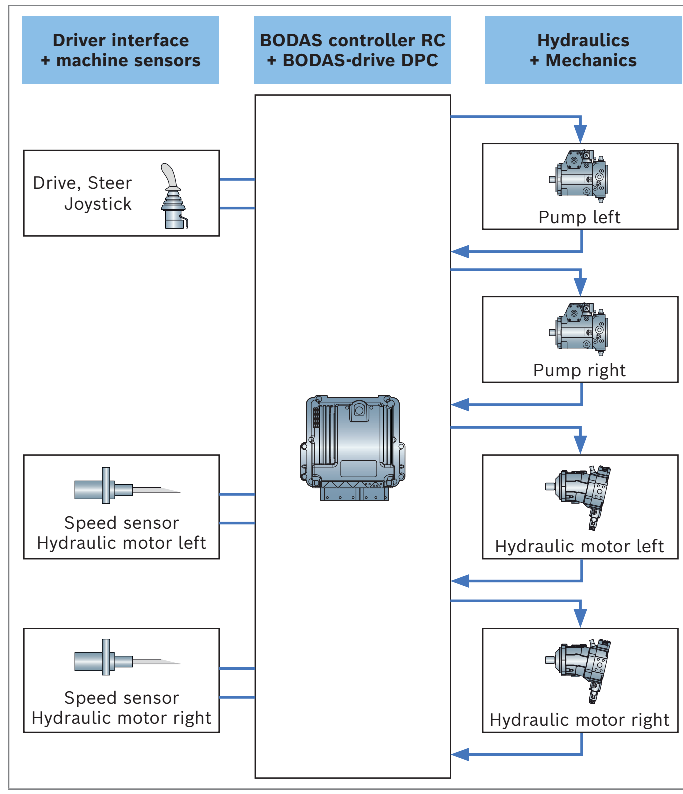
SF4: Safe direction

This safety function prevents the vehicle from propelling into wrong direction.
PFH value for the BODAS controller RC + BODAS-drive DPC subsystem depends on the connected sensors and the type of connection (such as analog or CAN...).



SF5: Safe steering

This safety function ensures that the vehicle steers in according with driver demand.
PFH value for the BODAS controller RC + BODAS-drive DPC subsystem depends on the connected sensors and the type of connection (such as analog or CAN...).



6.4 Safety functions for compact track loader

Safety Function

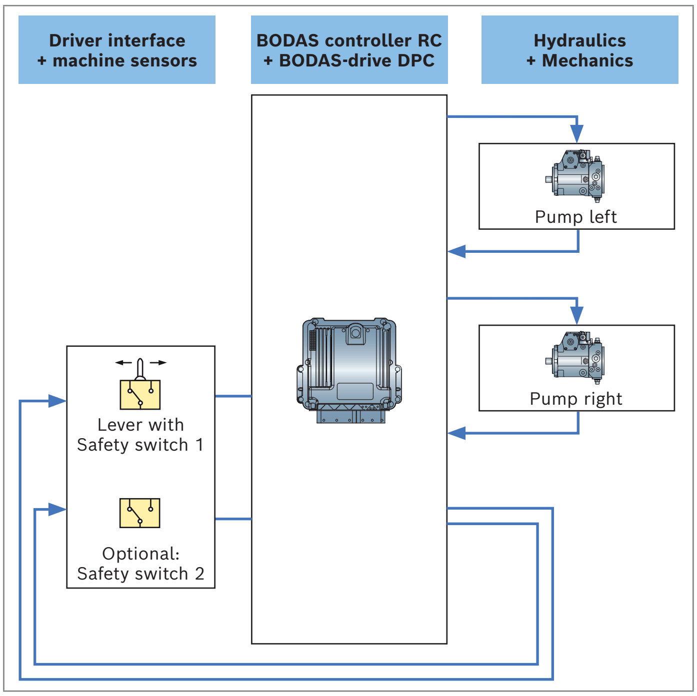
Function	**SF1 Emergency stop**[1]	**SF2 Safe standstill (driver inside)**	**SF3 Safe deceleration**	**SF4 Safe direction**	**SF5 Safe steering,**	**SF6 Safe acceleration**
in_drive: in_analog_drive (F2, F7)		x	x	x	x	x
in_drive: in_analog_steer (F2, F8)		x	x	x	x	x
in_CAN_Joystick: in_CAN_drive (F2, F7)		x	x	x	x	x
in_CAN_Joystick: in_CAN_steer (F2, F8)		x	x	x	x	x
in_Engine: in_accelerator (F13)			x			x
in_Engine: in_throttle (F12)			x			x
calc_Engine: calc_nominal_speed (F12)			x			x
calc_Engine: calc_pump_limit (F9, F15)			x			x
in_drive, action: in_switches (F9, F15)	x					
err_RC	x					
err_RC: error_reaction	x					
lib: state_and_drive_ramp (F9, F15)	x	x	x	x	x	x
lib: speed_controller (F17)	x	x	x	x	x	x
lib: steering (F10)	x	x	x	x	x	x
lib: steering_controller_LR (F10)	x	x	x	x	x	x
lib: steering controller separat (F10)	x	x	x	x	x	x
lib: pumps (F9, F15)	x	x	x	x	x	x
lib: brake (F9, F15)	x	x	x	x	x	x
out_drive (F4)	x	x	x	x	x	x

Relationship between functions and safety functions for compact track loader CTL
F: Function numbers, see detailed software documentation
lib: calc_DPC_VX.X.lib: calc_drive

1) Activated via “Safety Switch”, see system overview chapter 3.1

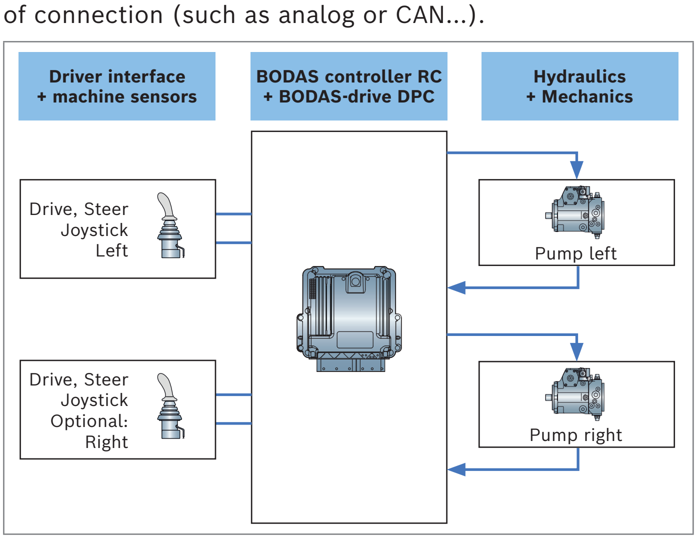
SF1: Emergency stop

This safety function stops the movement and inhibit vehicle movement from standstill if the safety lever is set down. The driver can be outside cabin.
PFH value for the BODAS controller RC + BODAS-drive DPC subsystem depends on the connected sensors for the safety switch(es).



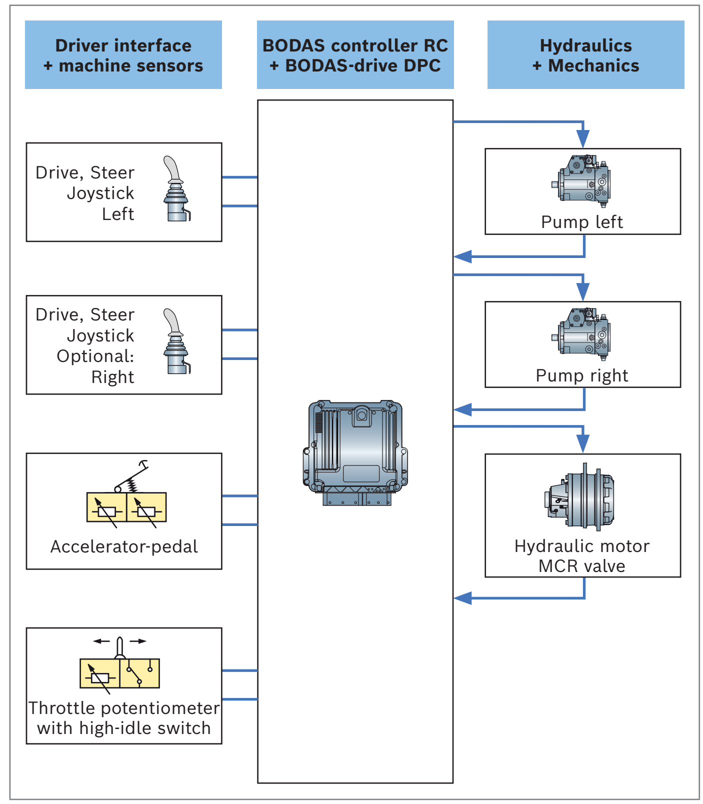
SF2: Safe standstill

This safety function ensures no unwanted self-propelled movement of the machine caused by an unwanted torque of the hydrostatic transmission in standstill condition if there is no driver command.
PFH value for the BODAS controller RC + BODAS-drive DPC subsystem depends on the connected sensors and the type of connection (such as analog or CAN...).



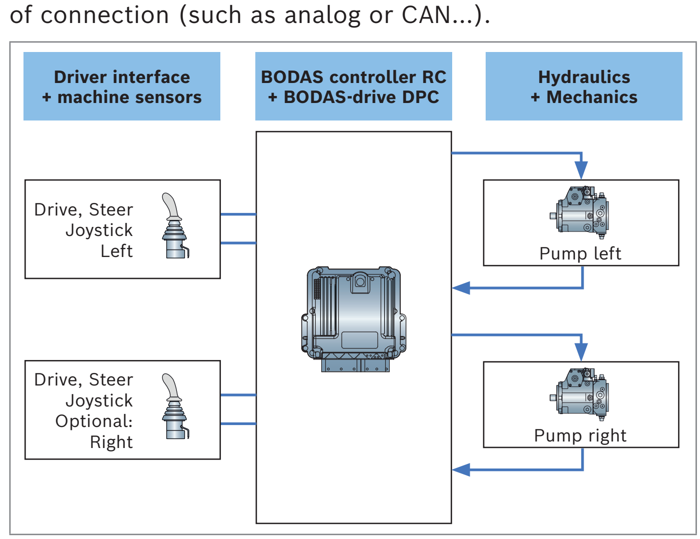
SF3: Safe deceleration

This safety function ensures that the hydrostatic transmission issues a braking torque when the driver demands it.
PFH value for the BODAS controller RC + BODAS-drive DPC subsystem depends on the connected sensors and the type of connection (such as analog or CAN...).



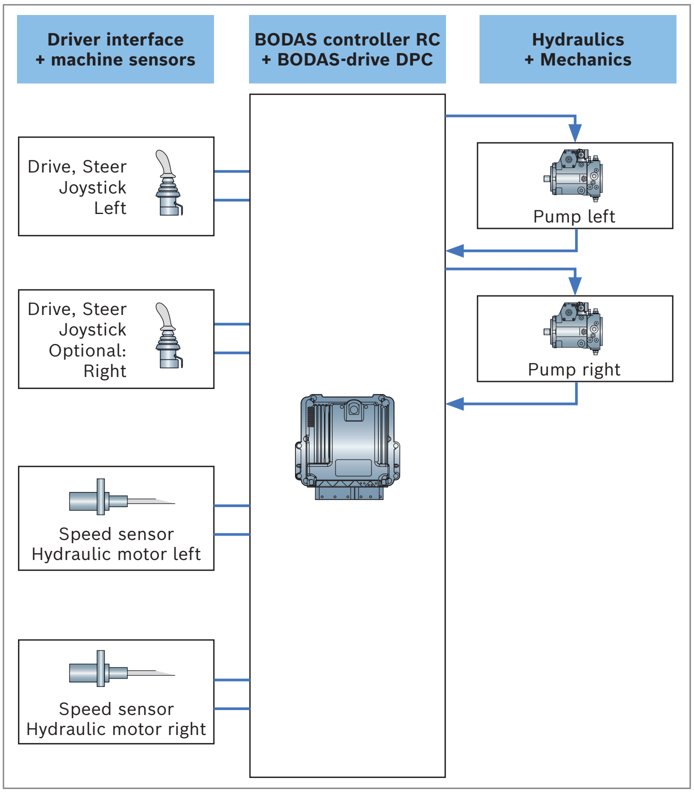
SF4: Safe direction

This safety function prevents the vehicle from propelling into wrong direction.
PFH value for the BODAS controller RC + BODAS-drive DPC subsystem depends on the connected sensors and the type of connection (such as analog or CAN...).



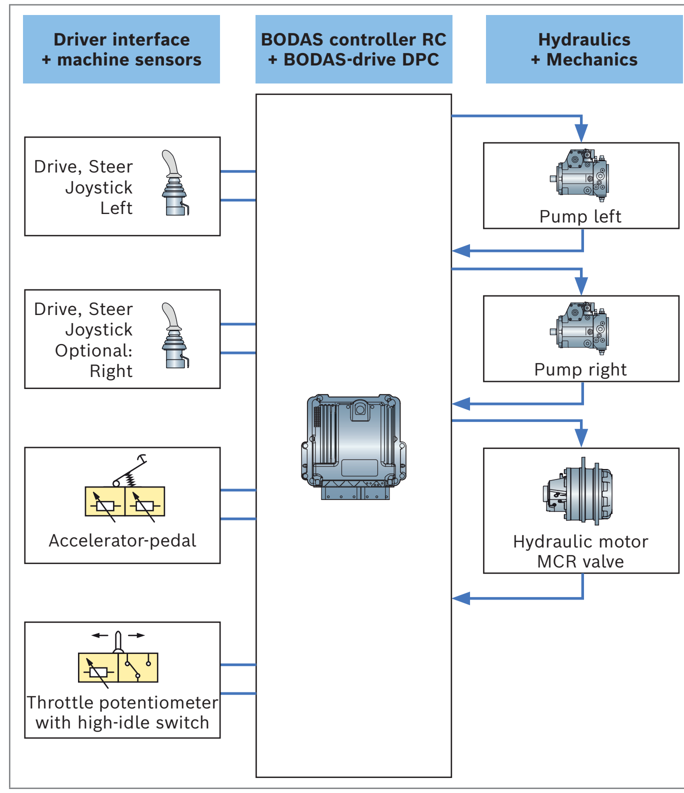
SF5: Safe steering

This safety function ensures that the vehicle steers in according with driver demand.
PFH value for the BODAS controller RC + BODAS-drive DPC subsystem depends on the connected sensors and the type of connection (such as analog or CAN...).



SF6: Safe acceleration

This safety function ensures, that the vehicle accelerates as the driver expects.
PFH value for the BODAS controller RC + BODAS-drive DPC subsystem depends on the connected sensors and the type of connection (such as analog or CAN...).



6.5 Example of using a BODAS-drive DPC safety function

This chapter uses an example based on SF2 Safe Standstill to show how to use the standard BODAS-drive DPC safety functions depending on the machine configuration.

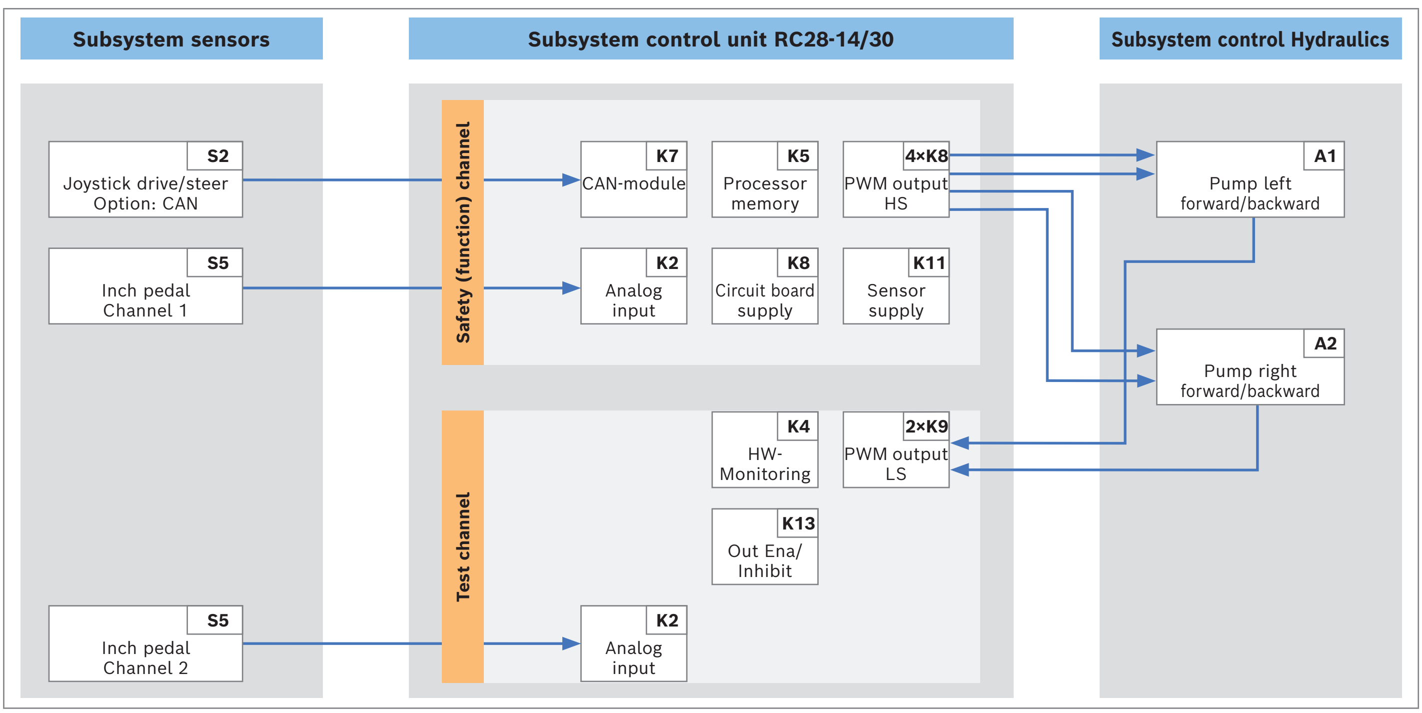
First step: Develop a block diagram

Develop the block diagram based on the machine configuration and desired functionality. The usable blocks of the BODAS controller RC are defined within the safety-relevant project instructions, Rexroth data sheet 95451-01-B of the choosen ECU. The appropriate blocks of the subsystem of sensors and hydraulics have to be chosen according to the machine configuration.

The example shows a block diagram for the safety function SF2, "Safe Standstill". It makes use of a Drive-Steer joystick and an inch pedal, an electrically controlled pump on the hydraulic side.

The block diagram shows the connection of all blocks relevant for this safety function. It also shows the chosen category, here category 2 with a function channel and a test channel.

▼ **SF2: Safe standstill, driver is inside cabin**



Second step: Calculate metrics

The metrics of $MTTF_D$ and DC value are calculated in the second step. Bosch Rexroth recommends using the SISTEMA tool for that purpose.

- SF Safe standstill
  - SB Sensors
    - CH Channel 1
      - BL S2 – Joystick drive/steer Option: CAN
      - BL S5 – Inch-pedal, channel 1
    - TE Test channel
      - BL S5 – Inch-pedal, channel 2
  - SB Control Unit RC
    - CH Channel 1
      - BL K7 – CAN bus
      - BL K11 – Sensor supply
      - BL K2 – Analog input
      - BL K5 – Processor and memory
      - BL K6 – Power supply unit and circuit board
      - BL K8 – PWM output highside pump left forward
      - BL K8 – PWM output highside pump left backward
      - BL K8 – PWM output highside pump right forward
      - BL K8 – PWM output highside pump right backward
    - TE Test channel
      - BL K2 – Analog input
      - BL K4 – Monitoring
      - BL K9 – PWM output lowside pump left
      - BL K9 – PWM output lowside pump right
      - BL K13 – INH (inhibit, pin 228)
  - SB Hydraulics
    - CH Channel 1
      - BL A1 – Pump control module left
      - BL A2 – Pump control module right

According to the BODAS controller RC + BODAS-drive DPC subsystem, the calculation for the full system has to be done based on the respective machine. The database of the used values can be found within the safety-relevant project planning instruction for BODAS controllers RC, Rexroth data sheet 95451-01-B. Since the sensor and system structures vary, a concrete calculation has to be done within each project. $MTTF_D$ values have to be requested from sensor manufacturers. Also, the machine-specific temperature profile has to be taken into account when performing the $MTTF_D$ value calculation

7 Project engineering and ordering information

7.1 Project engineering

The way from machine prototyping to serial production

- Thus to the fact that BODAS-drive DPC is a programming template a renaming of all BODAS-drive DPC files to project specific file names is mandatory.
- The scope of supply (as described above) must be applied and each controller has to be parameterized. Access to the parameters is password-protected. The password is part of the documents and tools container.
- The machine manufacturer must verify and validate his project in regard to the machine-specific requirements.

Important:

- The details in chapter 6, Functional safety, must be taken into account.
- Please take into account the Rexroth brochure 08511: “10 Steps to Performance Level”.

Required tools

- BODAS-service version 3.6 or higher; for the latest version see www.boschrexroth.com/mobile-electronics
- Microsoft Excel or equivalent for handling the Excel-to-BODAS configuration file and the approval test specification

Recommended tools

- SISTEMA, software tool for the application of the standard EN ISO 13849-1
- Vector CANalyzer Pro version 7.6 or higher

Required knowledge

- Participation on BODAS-design programming training
- Participation on BODAS-drive DPC Software training
- Knowing the controller RC12-10/30 or RC28-14/30
- Kowing the hydrostat
- Knowledge of functional safety EN ISO 13849
- For dozers, skid steer loaders: knowledge about ISO 15998, if applicable of ISO 25119

Project start guide

- Consideration of application specific standards for the relevant machine, if applicable
- Performance of a risk assessment by customer
- Identification of the safety functions
- Determination of the required performance level (PLr) for each safety function by customer required
- Safety concept should be written and SISTEMA calculations done, see chapter 6 “Safety functions”
- First program test of application should be made in the office with testboxes TB3 and not on the vehicle
- Configuration of CAN tool should be checked in the office before commissioning on the vehicle starts

7.2 Ordering code

01		02	03	04
ASopen	**–**	**DPC**	**E**	**10**

Type

01	Application software open for modifications	**ASopen**

Application

02	Dual path control for hydrostatic drivetrains for tracked mobile vehicles	**DPC**

Variant

03	Enhanced version	**E**

Release

04	Release number of the software	**10**

8 Valid standards and separate documentation

Document	
SAE J1939-21 December 2010	Data Link Layer
SAE J1939-71 May 2012	Vehicle Application Layer
Standard DIN ISO 13849-1: 2008-12 Standard DIN ISO 13849-2: 2012-10	Safety of machinery – Safety-related parts of control systems Part 1: General principles for design Part 2: Validation
Standard ISO 25119: 2010-06 Parts 1–4	Tractors and machinery for agriculture and forestry – Safety-related parts of control systems
Standard ISO/TS 15998-1: 2009-04-15	Earth-moving machinery ― Machine control systems (MCS) using electronic components ― Performance criteria and tests for functional safety
Technical Specification ISO/TS 15998-2	Earth-moving machinery ― Machine control systems (MCS) using electronic components ― Part 2: Use and application of ISO 15998
95204	BODAS Controller RC series 30 RC12-10, RC20-10, RC28-14
95451-01-B	Controller RC12-10, RC20-10, RC28-14 Safety-relevant project planning instruction
08511	10 Steps to Performance Level

▼ Compatible Rexroth products

Components	Data sheet	Relevant type code
Axial piston variable pump A4VG...EP and ET/40	92004	EP – Electric control, proportiona ET – Electric control, direct operated
Axial piston variable pump A4VG...EP/32	92003	EP – Electric control, proportional
Axial piston variable motor A6VM...EP/71	91610	EP – Electric control, proportional
Radial piston motor for wheel drives MCR-F	15198	
BODAS Speed sensor DSM	95132	
BODAS Speed sensor HDD	95135	
BODAS Speed sensor DSA_12	95133	
BODAS Temperature sensor fluid TSF	95180	
BODAS-service	95086	Version 3.6 or higher
BODAS measuring adapter MA8	95090	MA8/10
Joystick 4THED5 CAN	29696	
Joystick HG 405	95241	

9 Abbreviations

Abbreviation	Meaning
AgPL	Agriculture Performance Level
BODAS	Bosch Rexroth Design and Application System
BODAS-drive DPC	Dual Path Control Enhanced
CAN	Controller Area Network
CCF	Common Cause Failure
CCP	CAN Calibration Protocol
DC	Diagnostic Coverage
DIN	Deutsches Institut für Normung
DSA	Speed sensor type
DSM	Speed sensor type
ECU	Electronic control unit
EEC1	J1939 message from engine controller
EN	European Norm
EP	Proportional electric control
ET	Electric control, direct operated
HDD	Speed sensor type
HS	High-Side
IFA	Institute for work protection of the German statutory accident insurance
ISO	International Organization for Standardization
LS	Low-Side
MCR	Radial piston motor
$MTTF_D$	Mean Time To dangerous Failure
NPN	Tansistor type
PFH	Probability of dangerous failure per hour
PL	Performance Level
PLr	Required Performance Level
PWM	Pulse-width modulation
RC	Rexroth Controller
SAE	Society of Automotive Engineers
SF	Safety Function
SISTEMA	Safety integrity software tool for the evaluation of machine applications, produced by IFA
SRL	Software Requirement Level
TE	Test equipment
U_{Ign}	Voltage from ignition key
VTB	Virtual test box

10 Safety Instructions

- BODAS-drive DPC represents a safety element out of context. The machine manufacturer must verify whether it is the right product for the specific application.
- The machine manufacturer must perform a risk assessment.
- The required safety functions and performance levels must be fulfilled with the product in order to use BODAS-drive DPC in a specific application.
- The machine manufacturer bears responsibility for applying the valid safety standards at the machine level.
- The machine manufacturer is responsible for fulfilling all safety requirements at the drivetrain and machine level.
- The machine manufacturer is responsible for validating the machine-specific configuration of BODAS-drive eDA.
- Configurations of BODAS-drive eDA used for serial production must be validated.
- The proposed circuits do not imply any technical liability for the system on the part of Bosch Rexroth.
- Incorrect connections could cause unexpected signals at the outputs of the RC.
- Incorrect programming or parameter settings may create potential hazards while the machine is in operation.
- It is the responsibility of the machine manufacturer to identify hazards of this type in a hazard analysis and to bring them to the attention of the end user. Bosch Rexroth assumes no liability for dangers of this type.
- The application software must be installed and removed only by Bosch Rexroth or an authorized partner to preserve the warranty.
- It must be ensured that the vehicle is equipped with adequately dimensioned parking brakes.
- Make sure that the software configuration does not lead to safety-critical malfunctions of the complete system in the event of failure or malfunction. This type of system behavior may put life in danger and/or cause great damage to property.
- System developments, project specific code changes and installations and commissioning of electronic systems for controlling hydraulic drives must only be carried out by trained and experienced specialists who are sufficiently familiar with both the components used and the complete system.
- The machine may pose unforeseen hazards while commissioning and maintenance are carried out. Before commissioning the system, you must therefore ensure that the vehicle and the hydraulic system are in a safe condition.
- Make sure that nobody is in the machine's danger zone.
- No defective or incorrectly functioning components may be used. If the components should fail or demonstrate faulty operation, repairs must be performed immediately.
- The technical specifications and safety instructions of all involved components must be considered.
- The machine manufacturer must follow the valid standards and separate documentation (see chapter 8) when using the BODAS-drive DPC code in a project specific application.

Intended use

- The control unit is designed for use in mobile working machines provided.
  The machine has the following characteristics and is used on the following conditions and limitations:
  - Tracked vehicle or skid steer loader with wheels
  - Offroad using
  - Safe state: standstill
  - Safety functions developed according to the standards EN ISO 13849, ISO 15998, ISO 25119: maximum AgPL c, SW SRL1
  - Application derived from BODAS-drive DPC can use prepared safety functions based on the customer-specific risk assessment
  - A development process for application derived from BODAS-drive DPC is necessary to fulfill a safety standard
  - In any case, the requirements of the relevant safety standard must be fulfilled at the drivetrain and machine level
  - Depending on the safety requirements can the inherent or non inherent safety concept be used

- Operation of the control unit must generally occur within the operating ranges specified and released in this data sheet, particularly with regard to voltage, current, temperature, vibration, shock and other described environmental influences.
- Use outside of the specified and released boundary conditions may result in hazard to persons and/or cause damage to components which could result in subsequential damage to the mobile working machine.

Improper use

- Any use of the control unit other than as described under “Intended use” is considered to be improper.
- Use in explosive areas is not permissible.
- Use for passenger transportation is not permissible.
- Damage resulting from improper use and/or from unauthorized interference in the component not described in this data sheet render all warranty and liability claims void with respect to the manufacturer.

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